KU Leuven

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Your Thesis Title Here

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Abstract

Acknowledgements

Contents

Abstract **i**

Acknowledgements **iii**

Nomenclature **xi**

1 Introduction **1**

1.1 Background and Motivation 1

1.1.1 Musculoskeletal disorders and ergonomic work environments 1

1.1.2 Industrial motivation for ergonomic improvement 2

1.1.3 Physical human–robot interaction 2

1.1.4 Industrial manipulators for weight compensation 6

1.2 Problem Statement 6

1.3 Thesis Outline 6

2 Literature Review **7**

2.1 Theoretical Background 7

2.1.1 Example subsection 7

2.1.2 Another subsection 7

2.2 State of the Art 7

2.3 Gap Analysis 7

3 System Design and Implementation **9**

3.1 System Overview and Requirements 9

3.2 Mechanical Design 9

3.3 Hardware Architecture 9

3.3.1 CAN Bus Topology 9

3.3.2 Sensor Architecture 9

3.4 Sensor and Actuator Modelling 9

3.4.1 SET Spring Model 9

3.4.2 Motor and Transmission Model 9

3.5 Control Architecture 9

3.5.1 State Machine 9

3.5.2 Gravity Compensation and Floating Mode 9

3.5.3 Weighing Mode 9

3.5.4 Carrying Mode 9
3.5.5 Put-Down Mode 9
3.6 Software Implementation 9
3.6.1 Real-Time Architecture 9
3.6.2 Configuration 10

4 Validation and Results **11**
4.1 Experimental Setup 11
4.1.1 Hardware Configuration 11
4.1.2 Parameter Identification 11
4.2 Spring and Sensor Characterisation 11
4.3 Controller Performance 11
4.4 Assisted Load Handling Evaluation 11
4.5 Discussion 11

5 Conclusion and Future Work **13**
5.1 Conclusion 13
5.2 Future Work 13

References **15**

2D Manufacturing Drawings **17**
.1 CNC Part 1 17
.2 CNC Part 2 17
.3 CNC Part 3 17

Additional Experimental Results **19**

Source Code **21**

List of Figures

1.1 Human, robot, and shared workspace from Fraunhofer [6]. 3
1.2 Levels of cooperation between human worker and robot from Fraunhofer [6]. . 3

List of Tables

4.1 Example results table. 11

Nomenclature

Symbols

$\boldsymbol{q}$	Joint position vector	[rad]
K_s	Spring stiffness	[Nm/rad]

Abbreviations

CAN	Controller Area Network
SET	Series Elastic Transmission

Chapter 1

Introduction

1.1 Background and Motivation

1.1.1 Musculoskeletal disorders and ergonomic work environments

Non-ergonomic work environments place strain on the human body, which can cause painful complications and limit the worker's ability to perform tasks optimally. How these complications arise, which parts of the body are affected, and how they can be prevented are explained in this section.

The musculoskeletal system contains muscles, bones, joints and connective tissues like tendons and ligaments. Musculoskeletal disorders (MSDs) can therefore be defined as injuries of the muscles, nerves, tendons, joints, cartilage, and spinal discs. When these injuries are sustained or worsened by exposure to risk factors in the workplace, they are also known as work-related musculoskeletal disorders (WMSDs). These disorders can go from minor aches to severe pain and chronic illness requiring medical treatment, potentially preventing workers from performing physical tasks.

WMSDs develop over time due to non-ergonomic working conditions and can be categorized into two main groups:

- Back disorders related to handling loads
- Upper limb disorders, also known as repetitive strain injuries (RSIs)

MSDs can be caused by, for example:

- Twisting and bending movements while handling loads
- Repetitive or forceful movements
- Awkward and static postures
- Exposure to vibrating tools
- Prolonged sitting or standing in the same position

An increase in MSDs has been reported worldwide. This rise is partly attributed to improved recognition and reporting, but also to the tendency for work to be more segmented and specialized. This trend is driven by an increasingly competitive global economy, where fast-paced repetitive work environments are more widespread.

1.1.2 Industrial motivation for ergonomic improvement

Musculoskeletal disorders affect approximately one in three people worldwide. Within the European Union, MSDs represent the most costly health-related problems, affecting 45 million workers, many individuals are unable to continue working after the age of 60 due to these conditions, resulting in billions of euros in lost revenue and a substantial burden on the public healthcare system.

A declining physical condition can also lead to mental health issues such as depression or frustration, resulting in reduced job satisfaction and indirectly lower work performance, even when workers are still physically able to continue working.

The increase in education in the workforce, combined with higher expectations regarding job satisfaction, has reduced workers' tolerance for WMSDs. An ergonomic work environment can prevent the development of WMSDs and directly reduce both the cost to companies due to sick leave and the societal impact. Ergonomics can be defined as the study of work, where the goal is to fit tasks and jobs to workers' capabilities, rather then forcing workers to adapt to the job. An ergonomic work environment therefore implements task-specific strategies to prevent reduced work ability.

Reduced work ability can be categorized in absenteeism and presenteeism. Absenteeism refers to the absence from work due to injuries sustained on the job. Presenteeism describes the situation in which workers continue to attend their expected working duties despite existing injuries, often due to concerns about job security. In such cases, work is performed in a self–protective manner to reduce physical strain.

In Belgium, 2.5 million people were affected by MSDs in 2018 (approximately 62% of the Belgian working population in 2015), resulting in 3 billion euros in direct medical costs and 2 billion in work absenteeism, corresponding to 0.3% of Belgium's gross domestic product [1][2][3][4][5].

1.1.3 Physical human–robot interaction

There is a clear need to address reduced work efficiency resulting from physical injuries since this improves the quality of life of an increasingly aging workforce, but also to reduce the impact on social welfare systems and increase overall economic productivity. Industry has already adapted robotic solutions to achieve accuracy, speed, and repeatability that human workers cannot provide. Humans, however, can provide flexibility and have the capability to react to unforeseen circumstances. By increasing the degree of human participation, the strength and repeatability of robots is combined with the adaptability and decision making skills of humans. This schematic shown below illustrates the different workspaces in which robot and human operate.

Human–robot cooperation generally refers to the use of robots without safety fencing, also called cage–free robots. This in contrast to traditional robots which are stiff, heavy, and lack safety features that allow them to safely operate in the same space as humans, and are consequently only allowed to operate in closed of workspaces.

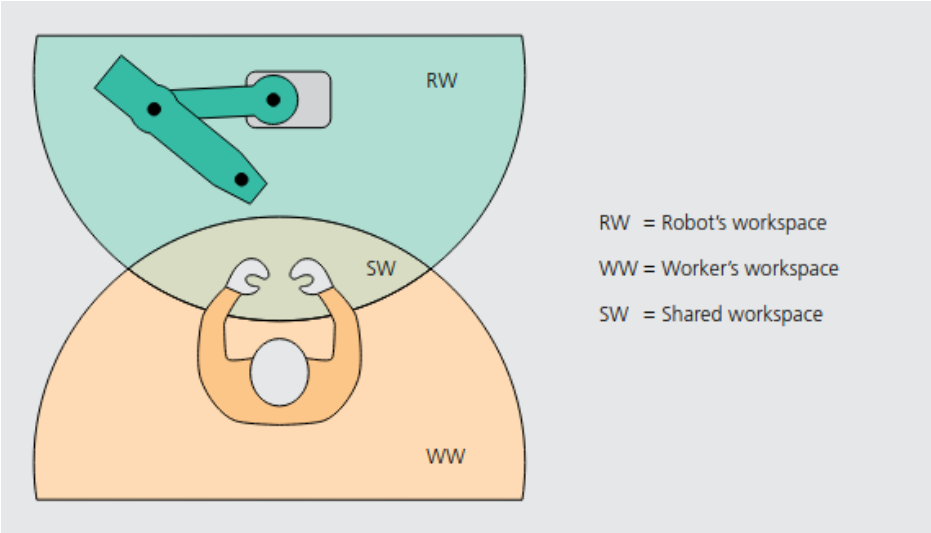

Figure 1.1: Human, robot, and shared workspace from Fraunhofer [6].

Human and robot can work together in different types of operation, as illustrated in Figure 1.2.

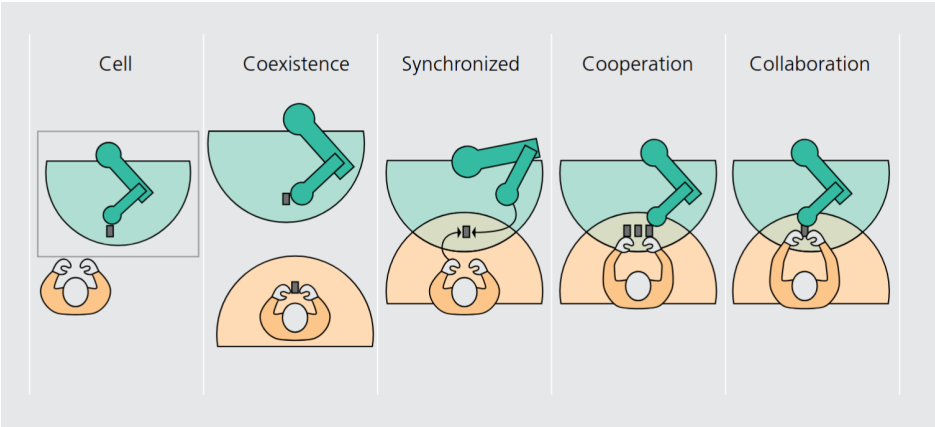

Figure 1.2: Levels of cooperation between human worker and robot from Fraunhofer [6].

From left to right an increasing level of interaction is shown.

- Cell: Robot operates in traditional cage, fenced of from human contact
- Coexistence: Human and cage–free robot work at the same time, but in different workspaces
- Synchronized: Human and cage–free robot work in the same workspace, but one at a time
- Cooperation: Human and cage–free robot have tasks in the same workspace at the same time, but to do not work together on the same task
- Collaboration: Human and cage–free robot work in the same workspace on the same task

The robotic testbench that was built in this thesis falls under coexistence and collaboration

depending on how it is implemented. This is clarified in chapter x?

Collaborative robots, defined as robots capable of performing collaborative tasks as described above, are widely used in industry. However, they are still primarily applied in traditional robotic operations such as pick–and–place tasks. In many cases, humans and robots remain either physically separated or operate in a coexistence mode, where no physical barrier is present but no direct interaction takes place. Such implementations are only economically viable for large batch sizes, since a whole production line needs to be tuned towards that specific robotic solution.

Cage–free robotic arms that share a workspace with a human worker are introduced to increase production flexibility. Interactive robotic solutions for small batches and varying product sizes has become a clear need since 70% of collaboration between human worker and robotic solutions are found in assembly applications. An industry survey reported that 86% of respondents were willing to invest in robotic solutions, indicating strong industrial interest. The primary motivations for introducing robotic arms in assembly environments are:

- Improving operational efficiency
- Improving ergonomics

Cage–free robotic solutions generally involve higher initial costs compared to traditional fenced systems due to safety certification requirements and conformity procedures to minimize the risk of injuries. Companies are however often willing to accept longer payback periods for implementing and testing company specific robotic solutions which increase ergonomics [6][7].

Normative ISO/TS 15066 covers the safety requirements for collaborative operations between human worker and robot. This normative classifies four different collaborative operations modes [8][9]:

- Safety–rated monitor stop
- Hand–guiding
- Power and force limiting
- Speed and separation monitoring

Safety–rated monitored stop The robot operates at its rated speed when a human is not present in the collaborative workspace. When a human enters the workspace, which can be detected by various means like laser scanners or light curtains, the robot undergoes a complete stop. As long as the human is in the workspace, the robot will not operate. Only by leaving the workspace and resetting the safety stop can operation continue.

Hand–guiding Hand–guided robotic applications are defined as a direct physical interaction between human and robot, where the human operator has full control over the robot motion. In this approach, a direct input device is located at or near the end–effector to measure input forces and simultaneously define the position of the human in the shared workspace. The robot must be in a monitored stop before hand–guiding can be activated via an enabling device. Safety requirements for this mode include force and or torque sensing at the guiding point, emergency stop accessible from the position where the robot is being guided, which is

typically at the end–effector, and a maximum speed limit during guided motion.

Power and force limiting This mode is the most commonly associated with collaborative robots. The robots' software and hardware are designed in such a way that any contact with a human cannot result in injury. Low robot mass, rounded surfaces and padding combined with active force and torque sensing, which limits contact forces with its environment below set thresholds depending on the body region of the human that can be in contact with the robot.

Speed and separation monitoring Robot and human can operate in the same workspace at the same time, but the position and speed of the human and robot relative to each other is constantly monitored to adjust the speed of the robot with the goal of constantly having a minimum dynamical separation distance at all times.

Hand–guiding is useful for packaging and assembly applications since it combines the physical strength of robotic arms, which can compensate the weight of components or help with moving relatively light but large components, with human intelligence.

ISO/TS 15066 specifies that the hand–guiding devices should be located at or near the robot end–effector. (not sure if its hand guiding or hand–guiding)

A direct consequence of this approach is that robotic applications with sensorised end–effectors can only react in a passive compliant manner when handled at a specific point on the robotic arm. Collisions or attempted handling of the robotic arm elsewhere will result in a safety stop of the robotic arm [10].

Existing robotic solutions that allow a human worker to hand–guide a robotic arm by grasping it wherever is most convenient are, to the best of the authors' knowledge, practically non–existent in assembly and packaging industries. When hand–guiding is discussed in literature, it is typically limited to sensorised end–effectors on conventional stiff robots. Collaborative robots that are adapted in industry are generally those without build in compliance. As stated before, collaborative robots are used for applications with little to no human contact where precision and repeatability is of more value than inherent compliance. [11][12].

This absence in current industrial applications is further supported by surveys of end users, system integrators and researchers, which indicate that collaborative robots are still primarily implemented as traditional stiff robots with limited or no direct human interaction but with the added value that less space is needed to operate these robots [13].

Collaborative robots such as the UR30 from Universal Robots, and the CR-35iB from FANUC can handle payloads up to 30kg and 50kg respectively. Hand–guiding is mainly used for point based teaching, and path based teaching. The user can intuitively move the robot to a new point in space and record that position. This can be done for a singular point, or for a full path, in this case the robot records the full movement from start to finish while a button is pressed. It is however possible to correctly configure collaborative robots for weight compensation, but software is needed for correct calibration of the load attached to the end–effector. Although rare, some collaborative robots can estimate payload weight and center of mass by performing predefined controlled motions, but multiple interactions with the robot software controller are needed making this approach time consuming. This function however is not found often, and is often an optional software package and is not available for

every collaborative robot, further proving that weight compensation of payloads of varying weights is mostly absent in industrial collaborative applications [14][15][16][17].

1.1.4 Industrial manipulators for weight compensation

Passive weight compensating manipulators are well established in industrial environments. By combining continuous force springs like coil springs or gas springs which can sometimes be adjusted, with load bearing joints, humans only have to provide a small force to move heavy objects. These are, as stated before, passive and can thus not be used for autonomous tasks. They are also pre-tuned for specific weights, and can only be used in a specific weight range. Pressurized air needs more maintenance then electric solutions and is also more expensive. I LL ADD TO THIS LATER

1.2 Problem Statement

1.3 Thesis Outline

Chapter 2

Literature Review

2.1 Theoretical Background

2.1.1 Example subsection

Example equation:

$$\tau = K_s \, \theta_s \tag{2.1}$$

2.1.2 Another subsection

2.2 State of the Art

2.3 Gap Analysis

Intrinsic safe robot Grasp wherever you want 3dof enough for packaging applications, no need for more dof

Chapter 3

System Design and Implementation

3.1 System Overview and Requirements

3.2 Mechanical Design

3.3 Hardware Architecture

3.3.1 CAN Bus Topology

3.3.2 Sensor Architecture

3.4 Sensor and Actuator Modelling

3.4.1 SET Spring Model

3.4.2 Motor and Transmission Model

3.5 Control Architecture

3.5.1 State Machine

3.5.2 Gravity Compensation and Floating Mode

3.5.3 Weighing Mode

3.5.4 Carrying Mode

3.5.5 Put-Down Mode

3.6 Software Implementation

3.6.1 Real-Time Architecture

Example code listing:

```
while (running) {
    clock_nanosleep(...);
    // your code here
}
```

Listing 3.1: Example listing.

3.6.2 Configuration

Chapter 4

Validation and Results

4.1 Experimental Setup

4.1.1 Hardware Configuration

4.1.2 Parameter Identification

4.2 Spring and Sensor Characterisation

4.3 Controller Performance

4.4 Assisted Load Handling Evaluation

4.5 Discussion

Example table:

Table 4.1: Example results table.

Metric	Target	Achieved
Value 1	X	Y
Value 2	X	Y

Chapter 5

Conclusion and Future Work

5.1 Conclusion

5.2 Future Work

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2D Manufacturing Drawings

.1 CNC Part 1

.2 CNC Part 2

.3 CNC Part 3

Additional Experimental Results

Source Code

https://github.com/username/repository